YANG, Yan

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Degrees

Aug 2017 – May 2019 M.Sc. Computer Science, Georgetown University (Washington, DC)

* **Projects**: shift-reduce semantic parser (deep reinforcement learning); neural morphology segmentation (Keras); N-gram; Perceptron; EM-based statistical machine translation * ***Supervisor:*** *Prof. Grace Hui Yang*

Sep 2002 – Jun 2006 B.Eng. Information Engineering, Southeast University (Nanjing, China)

* ***(2003)*** *Course Scholarship for Linear Algebra and Space Analytic Geometry* * *Admitted at age 16*

Other Education

(Aug 2024 - now) self-study:

- *The Principles of Quantum Mechanics*, P.A.M.Dirac (§1-20 / Ch 1-3; now on Ch 4)
- *Elements of Information Theory*, Tomas Cover (fully grasped Ch 1-8 and 14; touched on Ch 11, 13, and 15)
- MIT 18.102 *Introduction to Functional Analysis* (went over Lectures 1-8 and 14 Basic Hilbert Space Theory)
- *How Emotions Are Made: The Secret Life of the Brain*, Barrett, L.F. (Ch 1–10 of 11)

(Oct 2025) *Autumn School on fMRI*, @ IBRO & New York University, Abu Dhabi

- Exposure to experimental design and analysis workflows (Python-based), fMRI principles (HRF, preprocessing), and notebook-based modelling pipelines (e.g., GLM and second-order models).

(Sep 2023 - Nov 2024) Research stay, Computer Science, @ KAUST (Thuwal, Saudi Arabia)

(May 2022 – May 2023) *Probabilistic Graphical Model* (Part 2: 94/100; Part 1: 100/100) & *Computational Neuroscience* (99.57/100), @ Coursera

(Jun 2022) *Human Research (Social Behavior Research Investigators and Key Personnel Group, Basic)*, @ CITI Program & University of Nevada, Reno

(Jan 2021 – Aug 2022) Graduate research & coursework, Computer Science & Engineering, @ University of Nevada, Reno * ***Supervisors:*** *Prof. Mircea Nicolescu & Prof. George Bebis*

(Jan 2021 – Jun 2021) *Convolutional Neural Networks* (100/100) & *Mathematical Biostatistics Boot Camp* (100/100), @ Coursera

(Jan 2021) Training: *GRAD 701S – College Teaching*, @ University of Nevada, Reno

(Jul 2016) Gender studies summer intensive taught by Caroline Light (Harvard) with teaching assistants, in-person, Beijing; organized by Ruiji Institute; small-group, application-based, need-based tuition waiver

Research Output

Publications & Preprints:

<Cognitive State Modelling>

- Yang, Y. **(2026)**. Modeling Limits in Phenomenological Contexts – a Statement on Luo et al.'s *Improving Searcher Struggle Detection via the Reversal Theory*. *Journal of Motivation, Emotion, and Personality* 14 (2026), pp. 1-5. doi.org/10.12689/jmep.2026.1401

- Yang, Y. **(2026)**. From Satiation to Allostasis: Toward a Resource-Based Framework for Reversal Triggers in Reversal Theory. *Zenodo*. https://zenodo.org/records/19437171

- Jiyun, L., Yang, Y., Nayak, V., & Yang, G.H. **(2024)**. Improving Searcher Struggle Detection via the Reversal Theory. *Discover Computing* 27, 51. Springer Nature. doi.org/10.1007/s10791-024-09492-z

<Rule-Informed ML for Modeling>

• Yang, Y. **(2026)**. Loss Landscape Diagnosis for Gradient-Based Gray-Scott System Inversion: Disentangling the Roles of PINN Components. *AI4Physics@ICML 2026 Workshop*; under review by *TMLR*. https://arxiv.org/abs/2606.112

• Yang, Y. **(2026)**. EM-style Framework for Task-Driven Differentiable Synthesizer Optimization: Rationale, Design, and Backpropagation through Reaction–Diffusion System. *Zenodo*. https://zenodo.org/records/19397678

• Yang, Y., Bebis, G., & Nicolescu, M. **(2022)**. Two-Step Data Augmentation for Masked Face Detection and Recognition: Turning Fake Masks to Real. In Proceedings of *the 2nd International Conference on Image Processing and Vision Engineering – IMPROVE* (pp. 126-134). SciTePress. dx.doi.org/10.5220/0011037900003209

Methods, Software, & Research Infrastructures

(2025) Developed a Gray–Scott simulation codebase for backpropagation-based parameter inversion: https://github.com/Yan-Yang-bot/bp2renderer, which led to principled theoretical analyses and PINN design implications condensed into a poster at AI4Physics, ICML'26. *Independent Research @ self*

(2021 ~ 2023) Led prototyping of a CNN embedding visualization tool for cell images, including installation guides for the local-running prototype; later redesigned the system for commercial lab product integration, emphasizing scalability and incremental development. *R&D Intern / R&D Engineer @ Deepcell Inc.*

(2018) Revamped search engine's VR interface (hand model calibration & connection) for behavioral data collection and provided web technique support for its SIGIR conference demonstration, acknowledged by the corresponding author as authorship-worthy. *Research Assistant @ Georgetown University*

(2015) Led full-stack development of a weather monitoring application. (AngularJS, Flask, NodeJS, MySQL) *Full Stack Engineer @ ColorfulClouds Tech Co. Ltd*

Scientific and Societal Impact

(Oct 2026, upcoming) Poster Presenter @ Models of Consciousness 7, AMCS, University of Copenhagen
Yang, Y. Toward a Formal Account of Reversal Triggers: From Allostatic Regulation to Compositional-Distributional Semantics in Reversal Theory

(July 2026) Poster Presenter @ AI4Physics workshop, ICML 2026
Yang, Y. *Loss Landscape Diagnosis for Gradient-Based Gray-Scott System Inversion: Disentangling the Roles of PINN Components*

(2025 – 2026) Civic engagement on research integrity and constructive academic culture via public discourse.

(Sep – Oct 2025) Active Participant @ Models of Consciousness Conference 6
Contributed to consciousness modeling meta-discussions by introducing Gödel's incompleteness theorem.

(Apr 2025 – Mar 2026) LinkedIn creator (content relocated to SubStack: https://substack.com/@yy490)
Visualizations and essays on mathematical concepts.

(Sep 2022) Active Participant @ Models of Consciousness Conference 3
Raised discussion on temporal scales of psychological processes (short-term fluctuations vs long-term life-shaping trajectories) and related data collection; engaged in post-talk exchanges with faculty from CMU (CS, subconscious models) and Stanford (philosophy of mind, qualia).

(Apr 2022) Presenter @ IMPROVE (online due to COVID-19)
Yang, Y., Bebis, G., & Nicolescu, M. *Two-Step Data Augmentation for Masked Face Detection & Recognition*.

Language Skills

Mandarin – Native **English** – Fluent | Six-year US residency. Eight-year scientific/tech work including graduate schools in English. Led a collaborative 19th-century historical English translation project, coordinated the full-length translated text delivery, and conducted core editorial oversight.

Self-assessed: **French** – Basic beginner proficiency **Traditional Chinese** – Basic elementary proficiency

Chronological Work Experience[1]

Jul – Nov 2025 Reviewer @ IEEE Transactions on Affective Computing

Reviewed EEG-related ML manuscripts: corrected two serious conceptual misuses of cross-disciplinary terminology in a manuscript; identified deeper questions for empirical interpretation; and more.

Oct 2025 – Current Independent Researcher

• Diagnosed gradient-based inversion for the Gray-Scott reaction-diffusion system by backpropagating through an unrolled simulation and used this as an ablation to analyze PINN components — presented at the AI4Physics workshop @ ICML as a poster, with code released on GitHub.
• Published a commentary re-examining a prior co-authored paper applying Reversal Theory (RT) to search-behavior modeling; identified mismatches between the theory's phenomenological commitments and the paper's statistical operationalizations, and distinguished theory-inspired approximations with faithful implementations — in the Journal of Motivation, Emotion, and Personality.
• Participated in BAMΞ workshop on measurement (online), Models of Consciousness Conference 6 (Sapporo, Japan), and IBRO Autumn School on fMRI (NYU Abu Dhabi).
• One manuscript under review at TMLR, plus multiple research notes on Zenodo, including extending prior work on RT, proposing linkages between reversal triggers and Barrett's predictive allostasis framework, as well as between measurement disturbance in phenomenological states and quantum measurement principles.

Nov 2024 – Sep 2025 <Career Gap> *Freelance work, independent writing, medical recovery, and other activities listed in this CV within the same time period.*

Jul 2024 – Jan 2025 Reviewer @ IEEE Transactions on Neural Networks and Learning Systems

Sep 2023 – Nov 2024 Ph.D. Research Fellow @ KAUST (Thuwal, Saudi Arabia)[2]

• Investigated learning of dynamical system's parameters from data, a mathematical framework with direct connections to neuroscience modeling, which motivated my pursuit of the direction; this project aims to apply it to automate synthesis of diseased leaf images for agricultural classification models.
• Proposed ideas on Arabic Speech Recognition, including acoustic source separation and data augmentation in the spectrogram.

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[1] *Intentionally arranged chronologically to reflect the evolution of my research and professional interests*

[2] *Throughout my time at KAUST and after my departure, I sustained consistent independent study and self-directed research. Related outputs are documented on Zenodo, GitHub and SubStack, including a solo-authored paper submitted to AI4Physics@ICML 2026 (provided as the writing sample). Methodological and data integrity concerns raised at KAUST were repeatedly dismissed and deflected, undermining my ability to trust my own observations and rendering the working environment unproductive. I was removed from the lab and unable to secure a new supervisor, leading to withdrawal from the program. Some of these concerns (https://pubpeer.com/publications/7893CE90BA51FEEDCEF83F23B88FFA) are under journal retraction review; additional issues from unpublished work could not be documented via public peer review channels.*

May – Aug 2023 <Career Gap> *Road trip across half the US and personal life-planning milestone after a layoff and concurrently obtaining a Ph.D. offer.*

Oct 2022 – May 2023 Machine Learning R&D Engineer @ Deepcell Inc. (Mountain View, United States)

Led system design for remodeling the Neural Network embedding visualization tool, making it the human-in-the-loop decision point within the commercial Cluster-Based Sorting product: scientists in customer labs select sorting criteria in an interactive exploration session before the run proceeds.

Sep 2022 <Career Gap> *One-month gap between the OPT and H-1B work permits; attended Models of Consciousness Conference 3.*

May 2021 – Sep 2022 R&D Intern @ Deepcell Inc. (Mountain View, United States)

- *Trained GAN models to synthesize cell image data, augmenting training sets for downstream classifiers.*
- *Led development of a prototype CNN embedding visualization tool adopted by in-house data scientists and bioinformaticians for interactive clustering and correlation analysis; the tool became a direct precursor to the commercial Cluster-Based Sorting product built in the subsequent full-time role.*

Jun – Aug 2022 Volunteer Research Assistant @ Visual Perception Lab - Neuroscience, University of Nevada, Reno (Reno, United States)

Exploratory visit to assess fit for a neuroscience research placement: surveyed methods in Representational Similarity Analysis, Multivoxel Pattern Analysis, and fMRI-based face/object perception; discussed potential research directions with lab members; trained in BrainVoyager and human research ethics.

Jan – May 2022 Teaching Assistant @ University of Nevada, Reno (Reno, United States)

Recitation classes, grading, office hours, & exam proctorship, MATH 127 Precalculus II, including preparation of structured recitation slides and class exercises.

Sep 2021 – May 2022 Student Researcher @ University of Nevada, Reno (Reno, United States)

- *Engaged in a variety of course research projects, such as HPC log data analytics, medical data segmentation and classification, and adversarial learning used for data augmentation.*
- *Published Two-Step Data Augmentation for Masked Face Detection and Recognition.*

Jan – May 2021 Teaching & Research Assistant @ University of Nevada, Reno (Reno, United States)[3]

Grading and office hours, CS 320, Data Structure (C++).

Jun 2020 – Jan 2021 Member of Technical Staff @ Pure Storage (Bellevue, United States)

- *Implemented feature: import/restore volume snapshots from company disks into Kubernetes.*
- *Led the GitHub release process of Pure Service Orchestrator v 6.0.3.*

Jun 2019 – May 2020 <Career Gap> *Completed research postponed due to making up coursework after an in-semester medical leave; began job search, with placement delayed by the onset of COVID-19.*

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[3] *Original Ph.D. research funding was cut abruptly; the outgoing supervisor provided a data structure TAship as a transitional measure. During this period, I independently secured an industry internship and a math TAship and arranged temporary supervision under Profs. Bebis and Nicolescu while continuing to seek a stable research placement. The Visual Perception Lab participation was a concurrent exploratory attempt in that direction; it did not lead to a formal placement, as no engineering co-supervision was available.*

Jan 2018 – Sep 2019 Research Assistant @ Georgetown University (Washington, United States)

• *Led a critical step in anchoring the formalization of Reversal Theory's introspectively grounded motivational states—no prior math operationalization in the literature. The step grounded the formalization in RT's phenomenology and helped improve user struggle detection via search log data analysis.* (InfoSense Lab)
• *Revamped the search engine's VR interface for behavioral data collection, and provided web technique support for its SIGIR '18 conference demonstration, work acknowledged by the corresponding author as authorship-worthy.* (InfoSense Lab)
• *#MeToo Tweets Analytics (Georgetown course project, 3.4M tweets): applied LDA topic modeling (BoW & TF-IDF) to surface discourse themes (e.g., harassment/assault language, #MeToo historical framing, politicized subtopics); separately compiled GloVe-expanded occupational word-count analysis referencing U.S. Bureau of Labor Statistics categories* (Massive Data Institute)

Sep 2016 – Jul 2017 <Career Gap> ***Language training*** *and graduate-school applications; attended a small-group gender study summer intensive in Beijing taught by Caroline Light (Harvard), organized by Ruiji Institute.*

Sep 2013 – Aug 2016 Software/Web Engineer & Mentor @ various companies (Beijing, China)

Full-stack development across web apps, mobile mini-games, and desktop tools; mentored new employees on web development; stack included React-Redux, WebSocket, AngularJS, GoLang, Python, Redis database, NumPy, D3.js, C++

Mar – Sep 2013 <Career Gap> *Career transition from translation to software engineering via self-study in C++; independently developed a C++ minesweeper game.*

Jan 2009 – Sep 2013 Translator (EN-CN) @ Self-employed (Beijing, China)

• *Reorganized and led a collaborative Chinese translation project of* Women in the 19th Century *(Margaret Fuller, 1843) to completion; coordinated subsequent editorial review.*
• *Translated and proofread marketing materials, user/developer manuals, TV subtitles, and* software/game.
• *Published translations of science and technology news from media outlets such as Wired.com.*
• *Translated excerpts from Wendy Hollway's "Gender Difference and the Production of Subjectivity," in Changing the Subject: Psychology, Social Regulation and Subjectivity (Henriques et al., 1984), as independent study in feminist psychology and discourse analysis (2013)*

Jan 2008 – Jan 2009 Scientific Secretary @ Tongji University (Shanghai, China)

Jul 2006 – Dec 2008 <Career Gap> *Personal leave following undergraduate; concurrent with volunteer work below.*

Jul 2006 – Dec 2008 Psychology Peer Supporter (Volunteer) @ Ayaohelp.com (Remote, China)

• *Provided peer support drawing on self-study in psychology, spanning psychoanalytic, existentialist, humanist, and integral traditions, all sharing a resistance to reductive measurement of subjective states; deliberately supplemented the study with systematic academic foundations in experimental and general psychology (cognition, perception, learning, memory, research design, measurement, and validity).*
• *Reflective essays/poems on personal development were highlighted on front page by the site host, a Psychology professor at Beijing Normal University.*